TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Understanding of Compound-P -based regenerative mechanism and developing novel therapeutics for pulpitis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Small-Molecule Pharmaceuticals	Medical Bio	Bio-materials
Concept of Proposed Technology	**[Core Concept]** A paradigm shift in dental treatment from artificial material replacement to stem cell-based regenerative therapy for dentin. It involves developing **PNUT1**, a composite medical device consisting of the small molecule drug **Compound P (COMP)** and a collagen sponge carrier, which promotes the differentiation of Dental Pulp Stem Cells (DPSCs) into dentin-generating odontoblasts to regenerate lost dentin. **[Main Functions]** COMP accelerates the differentiation of DPSCs into odontoblasts and calcium accumulation. PNUT1 has achieved high-volume, high-density dentin regeneration in a mini-pig clean wound model. **[Potential Uses]** Application in **Vital Pulp Therapy (VPT)** for patients with pulpitis, shifting the prevailing standard of care (using inorganic cement) to a physiological regeneration therapy to restore the tooth's biological and functional integrity			

Rationale for Joint R&D Project	**[Need for Joint R&D]** While Compound P and PNUT1 showed superior dentin regeneration in a clean wound model, the exact target of Compound P in stem cells is still under investigation and the understanding of signaling network around the target will be critical for the future development. In the next step, its efficacy must be validated under **pulpitis conditions**, which is a prerequisite for human clinical trials. Pulpitis presents a varying spectrum of inflammation, necessitating the establishment of **inflammation control protocols** and **precision diagnostic technology** to maximize the preservation of viable pulp. **[Rationale for Collaboration]** StemDen specializes in applying COMP in DPSC culture, pulpitis model simulation, and optimizing PNUT1 formulation. Johns Hopkins University's strengths are in understanding the signaling network of COMP in stem cell of normal and pulpitis condition. This collaboration is essential to jointly understand the molecular mechanism of COMP in the application to a broad range of pulpitis patients by integrating StemDen's regenerative therapeutics with Johns Hopkins University's bioinformatics technology.
Key Words	1. dentin regeneration 2. dental pulp stem cell (DPSC) 3. odontoblast 4. pulpitis

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (3)	**End of the Project**	Level (4)
R&D Objectives	1. **Elucidate COMP Mechanism of Action and Identify Novel Targets**: Identify the molecular target and action mechanism of COMP in DPSCs to identify new drug targets for regenerative dentistry. 2. **Develop Pulpitis-Applicable PNUT1 Formulation**: Recapitulate pulpitis conditions in DPSC culture and assess the effect on COMP action, leading to the enhancement of PNUT1 formulation applicable in patients with pulpitis 3. **Develop Optimal PNUT1 Drug-Device Product**: Develop the PNUT1 drug-dental device master file for PMA filing for human clinical trials, and file PMA for the NIR-dye and imaging device.			
R&D Target Specifications	1. **COMP Mechanism**: Compare the efficacy of COMP and BMP-2 in promoting DPSC mineralization, aiming to establish **equal or superior efficacy of COMP compared to BMP-2** in a mini-pig model. 2. **PNUT1 Efficacy**: Establish efficacy by quantitatively analyzing **dentin regeneration and pulp preservation** using X-ray, micro-CT, and histological staining in rat and mini-pig **pulpitis models** after PNUT1 application 3. **Novel Analog Screening**: Screen **500 COMP analogues** for boosting DPSC mineralization and select optimized new COMP analogues in both normal and pulpitis conditions.			
R&D Plan	**[Research Approach]** The primary goal of this project is to shift the dental treatment paradigm from simple replacement to physiological regeneration by leveraging StemDen's proprietary compound, Compound P (COMP). The approach focuses on establishing an "Integrated Pulp Preservation and Regeneration Solution," specifically developing optimized combination formulations with inflammation-modulating drugs to ensure stable regenerative performance even under pulpitis conditions common in clinical practice based on deep understanding of molecular network of COMP signaling in stem cells. **[Key Tasks & Methodologies]** **1. Mechanism Elucidation & Pulpitis Modeling**			

- Identify the molecular target of COMP and analyze the response of DPSCs in pulpitis conditions.
- (Methodology)
  - ❍ Analysis of signaling pathways using Bulk and Single-cell RNA sequencing.
  - ❍ Establishment of rat and mini-pig animal models with induced pulpitis.

2: Optimization of Anti-inflammatory Combo-formulations & Efficacy Evaluation

- Enhance the PNUT1 formulation to maximize regenerative potential in pulpitis environments.
- (Methodology)
  - ❍ Determination of the optimal ratio between anti-inflammatory drugs (e.g., TNF-α inhibitors) and PNUT1, and validation of dentin formation efficacy.
  - ❍ Establishing the superior mineralization performance of COMP compared to BMP-2.

3. Preclinical Validation & Regulatory Preparation for Clinical Entry

- Final efficacy validation in pulpitis animal models and preparation for GMP manufacturing.
- (Methodology)
  - ❍ Quantitative assessment of dentin regeneration through Micro-CT and histological analysis.
  - ❍ Preparation of the Drug-Device Master File and regulatory filings to initiate human clinical trials.

[Expected Milestones & Timeline]

Phase	Key Milestones	Timeline
Phase 1	Identification of COMP molecular targets and establishment of standardized pulpitis models.	Jan 2026 - Dec 2026
Phase 2	Finalization of the optimal combination formulation and validation of superior efficacy.	Jan 2027 - Dec 2027
Phase 3	Acquisition of preclinical validation data and establishment of manufacturing processes for clinical trials.	Jan 2028 - Dec 2028

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

[Technical Contributions/Research Scope]

- **Target Validation of Compound P**: Elucidate COMP's molecular target and mechanism of action utilizing CellNet, gene expression, ATAC-seq, CRISPR-Cas9KO, etc.
- **Understanding Signaling Mechanism**: Using computational resources and combining biological data, analyze the signaling cascade centered at the target of COMP in stem cells.

[Resources and Specific Tasks]

- Apply computational analysis and CellNet in understanding the transcriptional and signaling network in controlling the differentiation of stem cells out of self-renewal and toward a specific lineage.

Expected R&R of Korean SME Partner	**[Technical Contributions/Development Tasks]** - **Pulpitis Model Simulation and Resolution**: Establish pulpitis models using DPSC culture and animal models (rat, mini-pig) and develop pulpitis-resolving solutions. - **COMP/PNUT1 Application in Pulpitis**: Evaluate COMP/PNUT1 efficacy in pulpitis conditions and derive optimal formulation (combination with anti-inflammatory drugs) **[Resource Commitments and Collaboration Efforts]** - Lead DPSC culture, COMP target validation, and discovery of new COMP derivatives - Prepare the pulpitis-resolving COMP/PNUT1 for GMP manufacturing and develop the PNUT1 drug-device master file

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**[Technological Achievements]** - **Identification of Novel COMP Targets:** Reveal critical and novel targets for therapeutic interventions in regenerative dentistry. - **Provision of Novel Protocols for Pulp Preservation**: Provide new pulpotomy protocols that maximize the preservation of viable pulp based on understanding pulpitis and infection/inflammation control. - **Development of Next-Generation COMP Derivatives:** Generate novel, rationally designed COMP analogues, strengthening StemDen's intellectual property portfolio. **[Commercialization Potential and Industrial Impact]** - **Paradigm Shift in Dental Care:** Validation of PNUT1 efficacy in pulpitis conditions will fundamentally change tooth decay treatment from age-old artificial replacement to physiological regeneration therapy - **Entry into Global Dental Market:** Preparation for human clinical trials and final regulatory approval PMA of PNUT1 will pave the way for entry into the global dental market (expected to reach 95 billion USD by 2032. - **Socioeconomic Benefit:** Improved quality of life and health for dental patients through regenerative therapy, leading to increased overall productivity and economic benefit for society.

6. References

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